



Grant Agreement no. 101120075

HORIZON-WIDERA-2022-TALENTS-03-01

ERA Talents

D3.3

Protocols and standard operating procedures on practical recommendations covering the data driven innovative method validation into the clinical practice

Embryo image-analysis workflow

WP3 — Intersectoral mobility for R&I translation towards ethically responsible innovations

Document information	Details
Submission date	30 September 2026
Lead beneficiary	UM
Dissemination level	PU — Public
Authors	Pending
Reviewers	Pending
Version / status	0.1 — Draft, pending review
Deliverable reference	D3.3 in the revised list; originally D3.4 in the proposal

Open-access web-based guidelines

<https://georgekazakis.github.io/nestor-sops/>

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Summary

Deliverable D3.3 describes a practical workflow for studying changes in the zona pellucida (ZP), inner cell mass (ICM) and trophectoderm (TE) during embryo development and hatching. The approach combines automated segmentation of time-lapse images, visual inspection and quantitative assessment using frame-level metrics.

Visual review was used to identify incorrectly segmented frames. Such frames were excluded from analysis; where too many frames were judged incorrect within an embryo, the whole embryo was excluded from analysis and metric extraction. Hatching was established visually, separately from the measurement CSV files.

The report provides a concise account of the approach and its quality-review procedure. It does not present detailed metric definitions, quantitative results or evidence of clinical readiness. The accompanying website provides the web-based version of the guidance.

Key conclusions

- Segmentation supported the study of structural and image-intensity changes over time.
- Visual review determined which frames and embryos were retained for analysis.
- The documented workflow provides a basis for further evaluation; it does not establish suitability for clinical decisions.

1. Introduction

1.1 Purpose

The purpose of D3.3 is to translate experience from NESTOR embryo image-analysis work into practical guidance on the analysis process, quality review and interpretation of its scope. It relates to WP3 and T3.1, which address translation challenges and the development of open-access guidelines in the form of protocols and standard operating procedures.

1.2 Scope

This report covers the embryo-analysis contribution within assisted reproductive technologies and AI/data-driven solutions for ART. The wider project also addresses personalised reproductive medicine, non-invasive prenatal testing and uterine health; these are not presented as case studies in this version. This report is distinct from D3.2 on interoperability and data management.

2. Practical analysis workflow

The research objective was to examine changes in ZP, ICM and TE during embryo development and hatching. The team trained a segmentation model on a separate computing workstation and used it within the embryo-analysis pipeline. The following stages summarise the approach described by the research team.

Stage	Approach
Segmentation	Identify ZP, ICM and TE in time-lapse frames using the trained model.
Quality review	Visually inspect each run and exclude incorrectly segmented frames or embryos.
Assessment of changes	Examine size, shape, ZP thickness and image intensity through visual review and quantitative metrics.
Record organisation	Store metrics in one CSV per embryo, with one row per time-lapse frame.
Hatching assessment	Establish hatching visually; hatching was not recorded in the measurement CSVs.

3. Visual quality-review procedure

The following procedure records the practice used in this work. It is a draft account for review, not an independently validated or universally applicable exclusion standard.

1. Visually inspect the segmentation results for each run.
2. Exclude frames that appear incorrectly segmented from the analysis.
3. Where too many frames within one embryo are judged incorrect, exclude that embryo entirely from analysis and metric extraction.

Embryo-level exclusion was based on qualitative visual judgement. No numerical cutoff is specified in this report. Interpretation therefore concerns the retained frames and embryos; this procedure does not establish performance across all originally processed images.

3.1 Proposed documentation improvements

For future use, a concise record of the model version, reviewed image sequences and reasons for exclusions would support traceability. Visual hatching observations could also be linked to the corresponding image sequence. These are proposed improvements, not claims that such records were completed in the reported work.

4. Embryo image-analysis example

The implemented workflow supports segmentation, inspection of overlays and videos, mask extraction and image-intensity visualisation. The team assessed changes both visually and through metrics recorded in embryo-specific CSV files. This report summarises that approach without reproducing private images or detailed measurement records.

The research implementation is maintained separately in the embryo-vision repository. The guidelines website explains the workflow and its limitations; it does not replace the software, research records or model evaluation. No quantitative findings from the measurement CSVs are asserted here.

5. Limitations and further validation

The documented work concerns analysis of embryo development and hatching. The qualitative exclusion procedure and separate visual identification of hatching are relevant when interpreting the retained data. Segmentation evaluation and any proposed clinical application require their own supporting evidence. This report does not establish suitability for embryo selection or treatment decisions.

6. Web-based guidelines

The accompanying website contains three pages: About the guidelines, Practical workflow, and Embryo image-analysis example. It provides an accessible web-based presentation of the deliverable content and can be updated as the documentation is reviewed.

<https://georgekazakis.github.io/nestor-sops/>

The website was accessible when this report was prepared on 28 September 2026. This document is version 0.1; the submission date is 30 September 2026. Authors and reviewers remain pending. Website publication does not itself constitute formal deliverable approval.

7. Conclusions

The activity combined automated segmentation, visual assessment and frame-level quantitative records to investigate embryo development and hatching. Its practical contribution is the description of an analysis workflow with visual quality review and exclusion of unsuitable outputs. This provides a basis for further documentation and evaluation within NESTOR while maintaining a clear distinction between research analysis and clinical validation.

References and project sources

NESTOR proposal SEP-210906745, 15 November 2022: ST2.2.3, printed page 22 (PDF page 93); T3.1, printed page 24 (PDF page 95); original deliverable table, printed page 28 (PDF page 99). The revised D3.3 identifier and submission date were provided for this report.

Research basis: the team's description of the activities and inspection of the embryo-vision documentation and analysis tools. The original D3.4 title is retained under the revised D3.3 identifier.

[NESTOR project website](#)

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